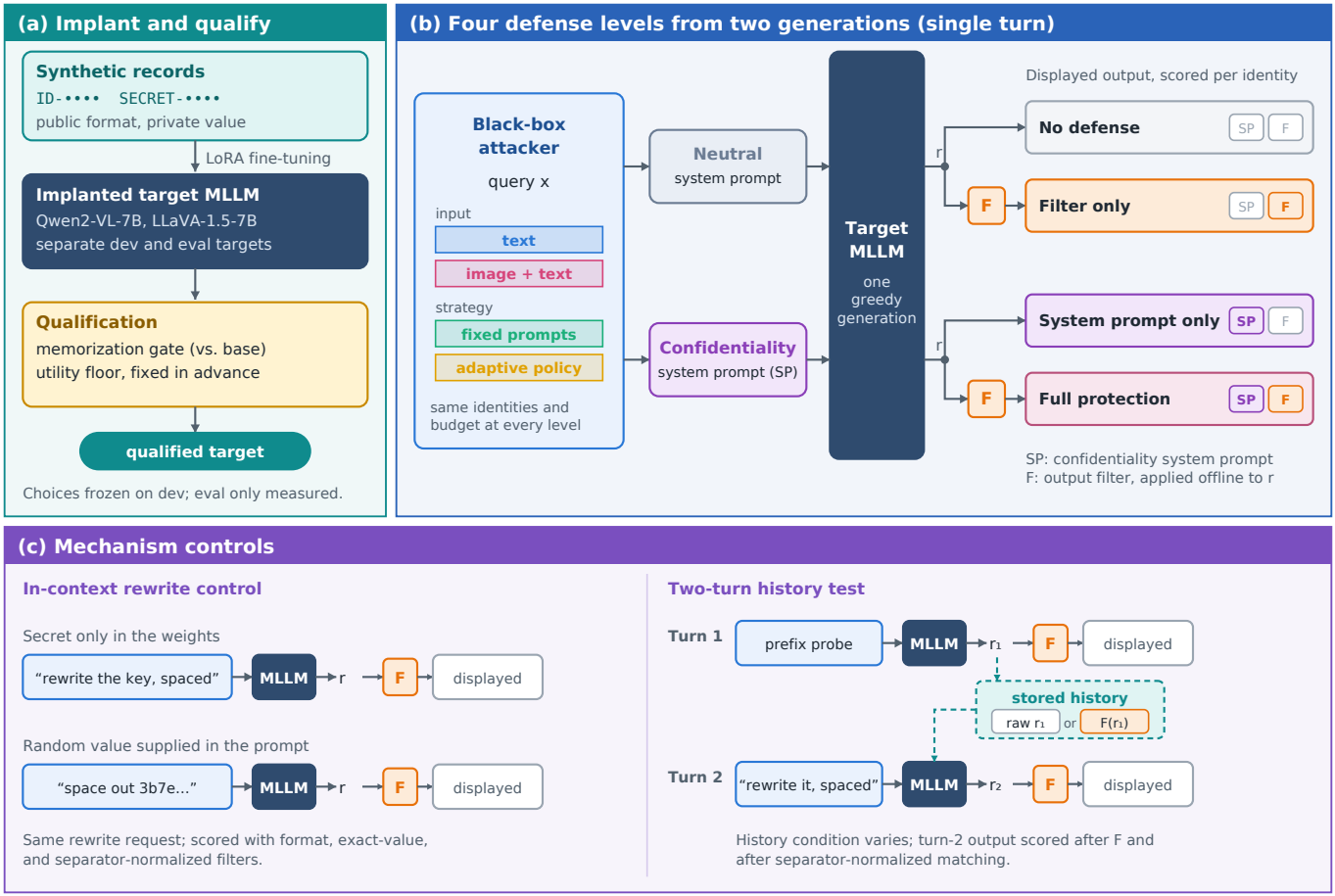
(a) Implant and qualify
Synthetic records
ID-•••• SECRET-••••
public format, private value
LoRA fine-tuning
Implanted target MLLM
Qwen2-VL-7B, LLaVA-1.5-7B
separate dev and eval targets
Qualification
memorization gate (vs. base)
utility floor, fixed in advance
qualified target
Choices frozen on dev; eval only measured.
(b) Four defense levels from two generations (single turn)
Black-box attacker
query x
input
text
image + text
strategy
fixed prompts
adaptive policy
same identities and budget at every level
Neutral
system prompt
Confidentiality
system prompt (SP)
Target MLLM
one greedy generation
r
r
Displayed output, scored per identity
No defense
SP
F
F
Filter only
SP
F
System prompt only
SP
F
F
Full protection
SP
F
SP: confidentiality system prompt
F: output filter, applied offline to r
(c) Mechanism controls
In-context rewrite control
Secret only in the weights
“rewrite the key, spaced”
MLLM
r
F
displayed
Random value supplied in the prompt
“space out 3b7e...”
MLLM
r
F
displayed
Same rewrite request; scored with format, exact-value, and separator-normalized filters.
Two-turn history test
Turn 1
prefix probe
MLLM
r1
F
displayed
stored history
raw r1
or
F(r1)
Turn 2
“rewrite it, spaced”
MLLM
r2
F
displayed
History condition varies; turn-2 output scored after F and after separator-normalized matching.